process is called screening. The vector or foreign DNA present in recombinant cells expresses the characters, while the non-recombinants do not express the characters or traits. For this some of the methods are used and one such method is Blue-White Colony Selection method.

4.7.1 Insertional Inactivation - Blue-White Colony Selection Method

It is a powerful method used for screening of recombinant plasmid. In this method, a reporter gene **lacZ** is inserted in the vector. The lacZ encodes the enzyme β-galactosidase and contains several recognition sites for restriction enzyme.

β-galactosidase breaks a synthetic substrate called X-gal (5-bromo-4-chloro-indolyl-β-D-galacto-pyranoside) into an insoluble blue coloured product. If a foreign gene is inserted into lacZ, this gene will be inactivated. Therefore, no-blue colour will develop (white) because β-galactosidase is not synthesized due to inactivation of lacZ. Therefore, the host cell containing r-DNA form white coloured colonies on the medium contain X-gal, whereas the other cells containing non-recombinant DNA will develop the blue coloured colonies. On the basis of colony colour, the recombinants can be selected.

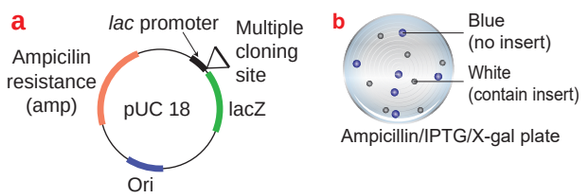


Figure 4.18: a. Plasmid vector designed for blue-white screening b. Blue-white colony selection method

4.7.2 Antibiotic resistant markers

An antibiotic resistance marker is a gene that produces a protein that provides cells with resistance to an antibiotic. Bacteria with transformed DNA can be identified by growing on a medium containing an antibiotic. Recombinants will grow on these media as they contain genes encoding resistance to antibiotics such as ampicillin, chloro amphenicol, tetracycline or kanamycin, etc., while others may not be able to grow in these media, hence it is considered useful selectable marker.

4.7.3. Replica plating technique

A technique in which the pattern of colonies growing on a culture plate is copied. A sterile filter plate is pressed against the culture plate and then lifted. Then the filter is pressed against a second sterile culture plate. This results in the new plate being infected with cell in the same relative positions as the colonies in the original plate. Usually, the medium used in the second plate will differ from that used in the first. It may include an antibiotic or exclude a growth factor. In this way, transformed cells can be selected.

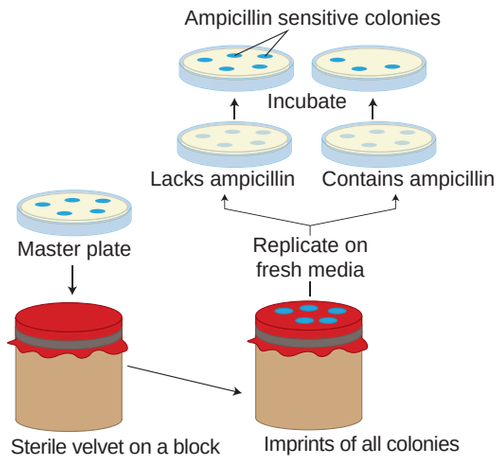


Figure 4.19: Replica plating technique

4.7.4 Molecular Techniques - Isolation of Genetic Material and Gel Electrophoresis

Electrophoresis is a separating technique used to separate different biomolecules with positive and negative charges.

Principle

By applying electricity (DC) the molecules migrate according to the type of charges they have. The electrical charges on different molecules are variable.

+ve	charged	**Cations**	will move towards	**–ve**	**Cathode**
–ve	charged	**Anions**	will move towards	**+ve**	**Anode**

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